CHAPTER 1: Limits, Continuity, and Differentiation

1.1 Limits and the Squeeze Theorem

A limit describes the value that a function $f(x)$ approaches as the input x approaches c.
Continuity is defined as the property where $\lim_{x \to c} f(x) = f(c)$ holds at an interior point c.
Indeterminate Form: A limit ratio such as $\frac{0}{0}$ whose limiting value requires algebraic simplification.

> **THEOREM 1.1 The Squeeze Theorem**
> If $h(x) \le f(x) \le g(x)$ for all x in an open interval containing c except possibly at c itself, and if $\lim_{x \to c} h(x) = L = \lim_{x \to c} g(x)$, then $\lim_{x \to c} f(x) = L$.

Special trigonometric limit: $\lim_{x \to 0} \frac{\sin x}{x} = 1$ underpins trigonometric differentiation.

1.2 Differentiation, Chain Rule, and Quotient Rule

The derivative of a function f at x is given by $f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$.
The Quotient Rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^{2}}$
The Chain Rule states that if $y = f(u)$ and $u = g(x)$ are differentiable, then $\frac{dy}{dx} = f'(g(x))g'(x)$.
The derivative of $\tan x$ is $\sec^{2} x$.
The derivative of $\ln x$ is $\frac{1}{x}$.
When the derivative $f'(x)$ is positive on an open interval, the function f increases across that interval.

CHAPTER 2: Definite Integrals and Infinite Series

2.1 Riemann Sums and the Fundamental Theorem of Calculus

Riemann Sum is defined as a finite approximation $\sum_{i=1}^{n} f(c_{i})\,\Delta x_{i}$ over a partition of $[a, b]$. Antiderivative refers to a differentiable function F satisfying $F'(x) = f(x)$ across an interval.

THEOREM 2.1 The Fundamental Theorem of Calculus
If a function f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on $[a, b]$, then $\int_{a}^{b} f(x)\,dx = F(b) - F(a)$.

Integration by Parts Formula: $\int u\,dv = uv - \int v\,du$

2.2 Convergence of Geometric and Taylor Series

Geometric Series: An infinite series $\sum_{n=0}^{\infty} ar^{n}$ with nonzero first term a and common ratio r.

THEOREM 2.2 Convergence of a Geometric Series
A geometric series with common ratio r converges to the sum $\frac{a}{1 - r}$ when $|r| < 1$, and the geometric series diverges whenever $|r| \ge 1$.

Taylor Series refers to the power series representation $\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x - c)^{n}$.
As the degree n of a Taylor polynomial increases, the truncation remainder decreases near the center c.

CHAPTER 3: Algorithms, Complexity, and Graph Theory

3.1 Asymptotic Complexity, Recurrences, and Divide-and-Conquer

Asymptotic Complexity is defined as the growth rate of an algorithm running time $T(n)$ as input size $n \to \infty$.
Merge sort runs in $O(n \log n)$ time in the worst case by recursively sorting two subarrays of size $n/2$.
The recurrence for merge sort is $T(n) = 2T(n/2) + O(n)$.
The running time of binary search is $O(\log n)$.

> **THEOREM 3.1 The Master Theorem**
> For a divide-and-conquer recurrence $T(n) = aT(n/b) + \Theta(n^{d})$ with constants $a \ge 1$ and $b > 1$, if $d = \log_{b} a$ then the asymptotic solution satisfies $T(n) = \Theta(n^{d} \log n)$.

Iterative binary search maintains midpoint $m = \lfloor (low + high) / 2 \rfloor$ and halves $[low, high]$.

3.2 Graph Traversals and Shortest-Path Algorithms

Directed Acyclic Graph: A directed graph with no directed cycles that admits a linear topological ordering.
Binary Heap refers to a complete binary tree satisfying the heap-order property at every internal node.
Dijkstra’s algorithm runs in $O((V + E) \log V)$ time on weighted graphs with nonnegative edge weights.
When graph edge density increases toward V^{2}, adjacency-matrix lookup latency decreases relative to lists.